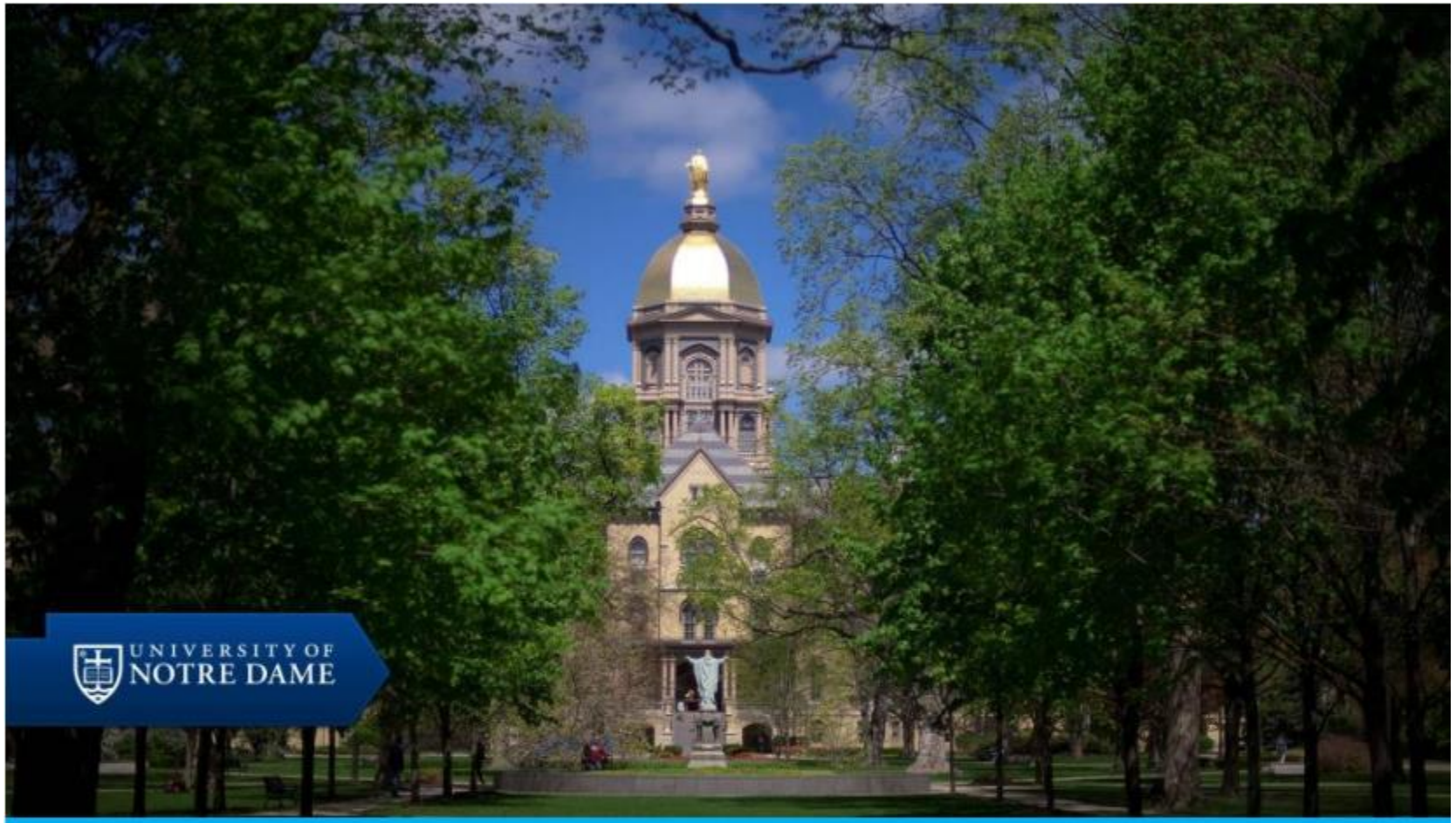


NOTRE DAME SUMMER WORKSHOP IN TOPOLOGY/GEOMETRY - 2018

Part I: A first glimpse - July 30 - August 1 - The first 2.5 days of the workshop will have two or three short courses assuming very little mathematical background. Potential topics are: "What is a manifold?" or "What is curvature?". A typical participant in this part would have taken a Calculus sequence as well as a few other math courses such as Linear Algebra.

Part II: Advanced topics - August 1 – 3 -The last 2.5 days of the workshop will consist of talks outlining more advanced topics of current research interest. In addition there will be a poster session where undergraduates can present their research. Some background in topology or geometry will be assumed from the participants. **APPLY FOR PART I OR II OR BOTH!**

Guest lecturers from 2017 (part I):

Arlo Caine: Curvature

Amanda Serenevy: Mathematical origami

Candice Price: Knot theory

Monday

8:45	*Opening remarks (Matt Gursky)*
9 - 10:30	Arlo Caine, I
11 - 12:30	Arlo Caine, II
12:30 - 2	*Lunch break*
2 - 3:30	Candice Price, I
4 - 5:30	Candice Price, II
6	*Cookout*

Tuesday

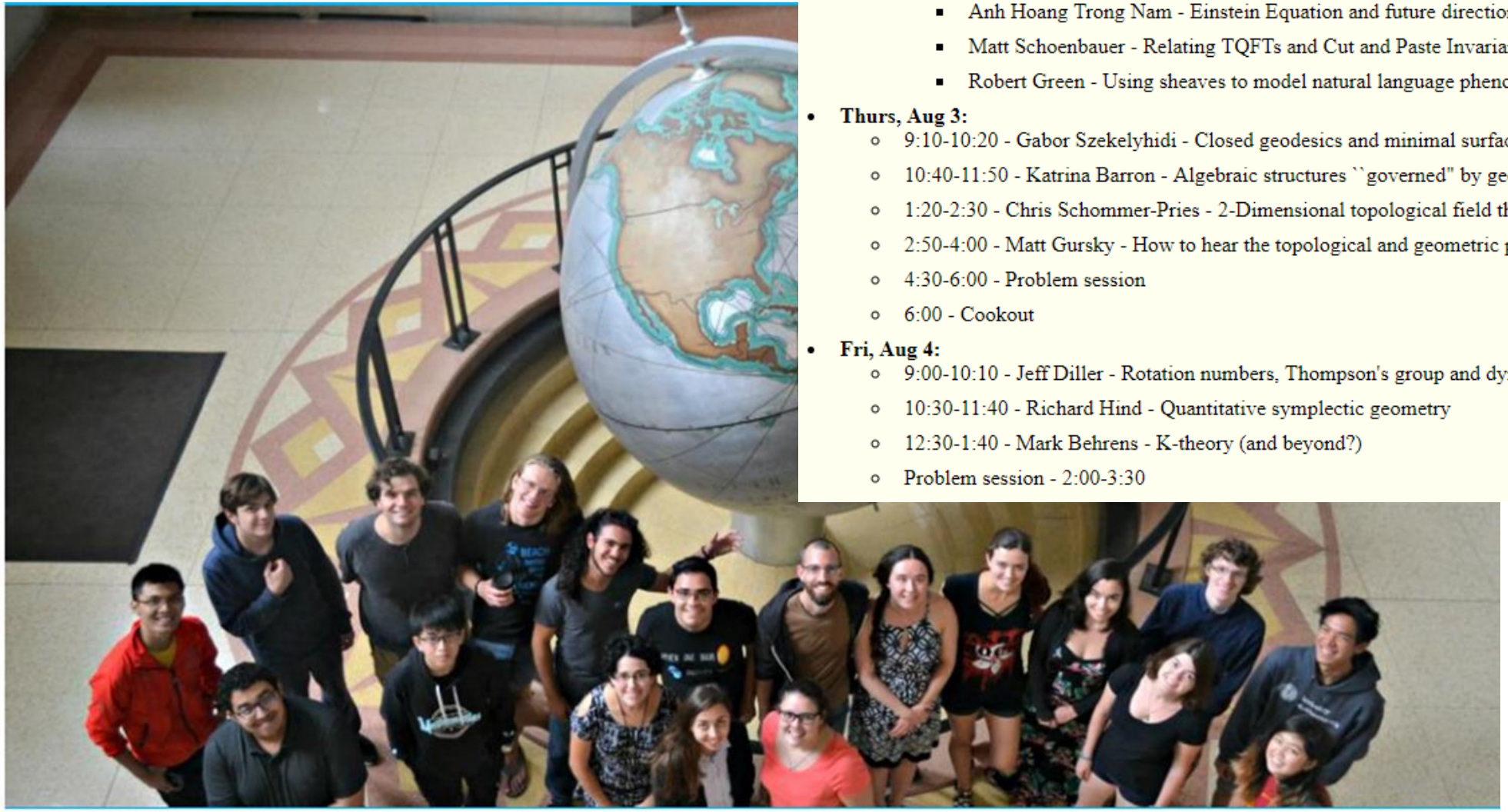
9 - 10:30	Candice Price, III
11 - 12:30	Candice Price, IV
12:30 - 2	*Lunch break*
2 - 3:30	Arlo Caine, III
4 - 5:30	Arlo Caine, IV
5:30	*Closing remarks (Matt Gursky)*

Wednesday

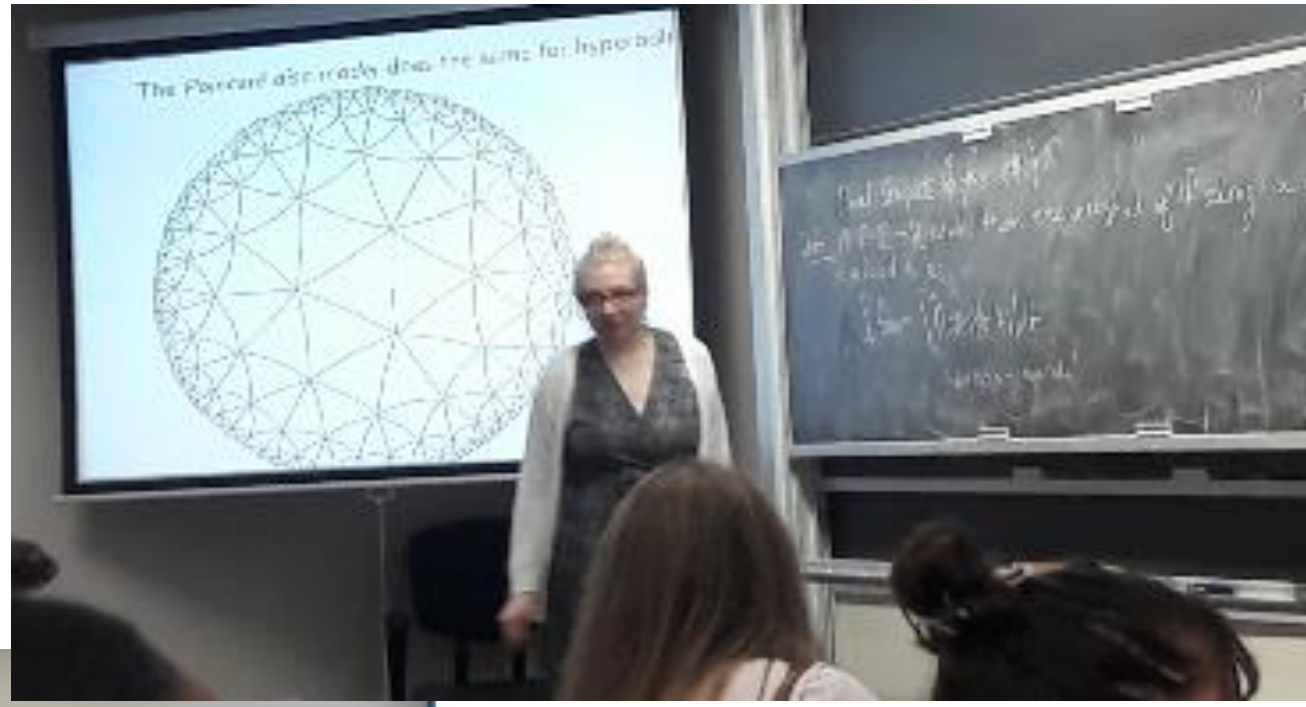
9 - 10:30	Amanda Serenevy, I
11 - 12:30	Amanda Serenevy, II

- **Wed, Aug 2:**
  - 1:20-2:30 - Stephan Stolz - Homology of surfaces
  - 2:50-4:00 - Andy Putman - Covers and simple closed curves
  - 4:30 - Undergraduate presentations:
    - Francesca Falzon - On the Complexity of the Rooted Triples Problem
    - Siddarth Kannan - Frobenius distributions in short intervals for elliptic curves
    - Claudia Sofia Carillo Vazquez - The Sortability of Graphs and Matrices under Context Directed Swaps.
    - Eric Chen - Geometry of the Affine Grassmannian
    - Matt Schoenbauer - Relating TQFTs and Cut and Paste Invariants
    - Anh Hoang Trong Nam - Einstein Equation and future directions in general relativity
    - Matt Schoenbauer - Relating TQFTs and Cut and Paste Invariants
    - Robert Green - Using sheaves to model natural language phenomena
- **Thurs, Aug 3:**
  - 9:10-10:20 - Gabor Szekelyhidi - Closed geodesics and minimal surfaces
  - 10:40-11:50 - Katrina Barron - Algebraic structures ``governed'' by geometric surfaces and applications to string theory
  - 1:20-2:30 - Chris Schommer-Pries - 2-Dimensional topological field theories and commutative frobenius algebra
  - 2:50-4:00 - Matt Gursky - How to hear the topological and geometric properties of a manifold
  - 4:30-6:00 - Problem session
  - 6:00 - Cookout
- **Fri, Aug 4:**
  - 9:00-10:10 - Jeff Diller - Rotation numbers, Thompson's group and dynamics of plane rational maps
  - 10:30-11:40 - Richard Hind - Quantitative symplectic geometry
  - 12:30-1:40 - Mark Behrens - K-theory (and beyond?)
  - Problem session - 2:00-3:30

Participants in Part II - 2017



Part I 2018: Guest speaker Rosemary Guzman:
Hyperbolic geometry and tilings



Part I group photo 2018

Part I Schedule

All talks are in Hayes-Healy Hall (connected to Hurley Hall).

Mon (229 HH)

9 - 10:30	Metric spaces, topological spaces, surfaces, and manifolds (Mark Behrens)
11 - 12:30	Problem session
12:30 - 2	*Lunch break*
2 - 3:30	Geometries, geodesics, isometries, and tilings (Rosemary Guzman)
4 - 5:30	Problem session
6	*Cookout*

Tues (231HH)

9 - 10:30	Classification of surface and Euler Characteristic (Mark Behrens)
11 - 12:30	Problem session
12:30 - 2	*Lunch break*
2 - 3:30	Mobius transformations, hyperbolic tilings, and their identification spaces (Rosemary Guzman)
4 - 5:30	Problem session
5:30	*Grad school application panel*

Wed (231 HH)

9 - 10:30	Knots (Mark Behrens)
11 - 12:30	Vistas (Rosemary Guzman)

Part II: Advanced topics

August 1 - 3

The last two and a half days of the workshop will consist of talks outlining more advanced topics of current research interest. In addition there will be an undergraduate presentation session where undergraduates can present their research. Some background in topology or geometry will be assumed from the participants.

Faculty talks and Short Research Reports are in *231 Hayes-Healy*
Coffee breaks are in the Math Department Lounge
Problem sessions will be in two rooms, *231 Hayes Healy* and *129 Hayes Healy*.

Schedule

Wednesday 8/1

1:30 pm	**Liviu Nicolaescu**	The many faces of the Gauss-Bonnet theorem
2:30 pm	Coffee break	
3:00 pm	**Richard Hind**	Quantitative symplectic geometry
4:15 pm	Short Research Reports	**S. Cetin, J. Erickson, M. Figging, B. Gales, S. Haney**
6:00 pm	Cookout	

Thursday 8/2

9:30 am	**Gabor Szekelyhidi**	Closed geodesics
10:30 am	Problem Session	
12:15-2:00 pm	Lunch break	
2:00 pm	**Katrina Barron**	Algebraic structures "governed" by geometric surfaces and applications to string theory
3:00 pm	Coffee break	
3:15 pm	Short Research Reports	**C. Hatton, I. Marchenka, S. Slaoui**
4:15 pm	**Pavel Mnev**	Moduli spaces of flat connections

Friday 8/3

9:30 am	**Qing Han**	The isometric embedding of surfaces in 3-dimensional Euclidean space
10:30 am	Coffee break	
11:00 am	**Mark Behrens**	Homotopy groups
12:00 - 1:30 pm	Lunch break	
1:30 - 3:30 pm	Problem Session	